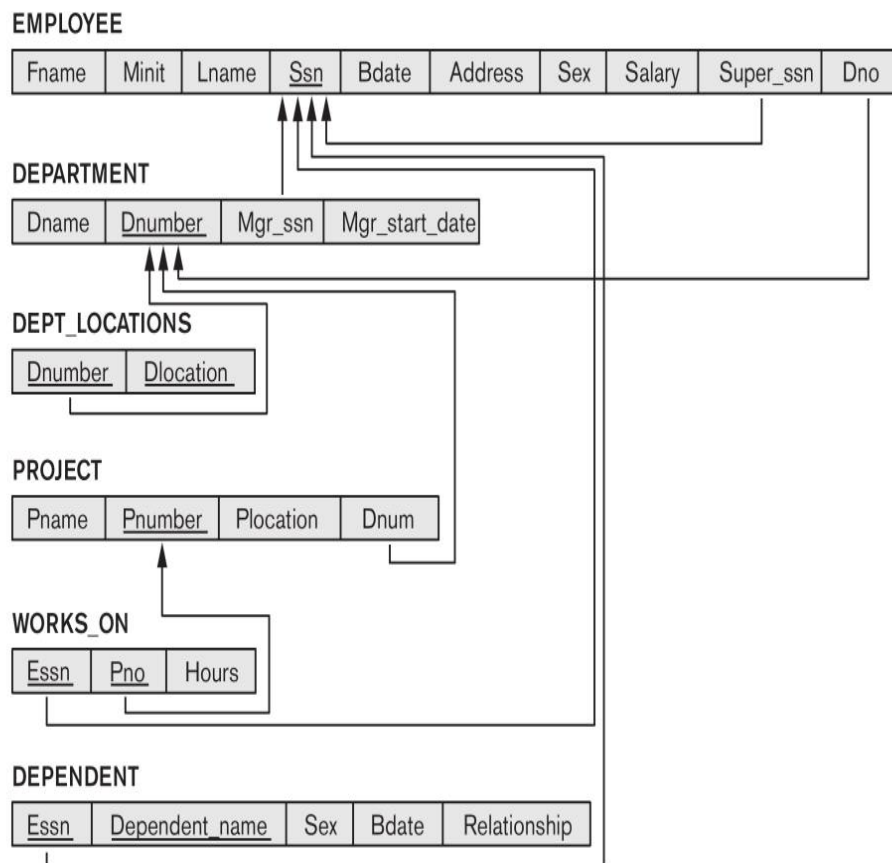


Sheet_7

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Examiners	: Dr. Tarek salem Eng : Eman, Habiba,Basant	Duration	2 hours
Course/Code	: Database Management Systems / CSE112		

Using the Company Database schema you created (which includes the following tables: EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, PROJECT, WORKS_ON, and DEPENDENT)

Resulting Relational Schema



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1. Retrieve all first names of employees:

```
SELECT Fname  
  
FROM EMPLOYEE
```

2. Retrieve all project numbers.

```
SELECT Pnumber  
  
FROM PROJECT;
```

3. Retrieve the distinct first names of employees.

```
SELECT DISTINCT Fname  
  
FROM EMPLOYEE;
```

4. Retrieve the distinct relationships of dependents.

```
SELECT DISTINCT Relationship  
  
FROM DEPENDENT;
```

5. Retrieve all locations, whether they are department locations or project locations (removing duplicates).

```
SELECT Dlocation FROM DEPT_LOCATIONS  
  
UNION  
  
SELECT Plocation FROM PROJECT
```

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6. Retrieve all department numbers that have employees or have locations (removing duplicates)

SELECT Dno **FROM** EMPLOYEE

UNION

SELECT Dnumber **FROM** DEPT_LOCATIONS;

EXCEPT` (Difference) Queries (Rows in the First Query but Not the Second):

7. Retrieve department numbers that have locations but do not have projects (the specific keyword `EXCEPT` might vary across database systems).

SELECT Dnumber **FROM** DEPT_LOCATIONS

EXCEPT

SELECT Dnum **FROM** PROJECT;

8. Retrieve department numbers where employees work but do not have locations (the specific keyword `EXCEPT` might vary across database systems).

SELECT Dno **FROM** EMPLOYEE

EXCEPT

SELECT Dnumber **FROM** DEPT_LOCATIONS

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INTERSECT` Queries (Rows Present in Both Queries):

9. Retrieve department numbers that have locations and have projects (the specific keyword `INTERSECT` might vary across database systems).

```
SELECT Dnumber FROM DEPT_LOCATIONS
```

```
INTERSECT
```

```
SELECT Dnum FROM PROJECT;
```

10. Retrieve department numbers where employees work and have locations (the specific keyword `INTERSECT` might vary across database systems).

```
SELECT Dno FROM EMPLOYEE
```

```
INTERSECT
```

```
SELECT Dnumber FROM DEPT_LOCATIONS;
```

`UNION ALL` Queries (Keeps Duplicate Rows):

11. Retrieve all locations, whether they are department locations or project locations (keeping duplicates).

```
SELECT Dlocation FROM DEPT_LOCATIONS
```

```
UNION ALL
```

```
SELECT Plocation FROM PROJECT;
```

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12. Retrieve all department numbers that have employees or have locations (keeping duplicates).

```
SELECT Dno FROM EMPLOYEE
```

```
UNION ALL
```

```
SELECT Dnumber FROM DEPT_LOCATIONS;
```

‘LIKE’ Queries (Pattern Matching):

13. Retrieve the first and last names of employees whose first name starts with 'J'.

```
SELECT Fname, Lname
```

```
FROM EMPLOYEE
```

```
WHERE Fname LIKE 'J%';
```

14. Retrieve the names of projects that contain the word 'Product'.

```
SELECT Pname
```

```
FROM PROJECT
```

```
WHERE Pname LIKE '%Product%';
```

15. Retrieve the names of dependents that end with the letter 'e'.

```
SELECT Dependent_name
```

```
FROM DEPENDENT
```

```
WHERE Dependent_name LIKE '%e';
```

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16. Retrieve the first names of employees where the second letter of their first name is 'o'.

```
SELECT Fname  
FROM EMPLOYEE  
WHERE Fname LIKE '_o%';
```

17. Retrieve the department locations that consist of 7 characters.

```
SELECT Dlocation  
FROM DEPT_LOCATIONS  
WHERE Dlocation LIKE '_____';
```

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Advanced Queries

Retrieve the first name, last name, and salary of employees, ordered by salary in ascending order.

```
SQL
SELECT Fname, Lname, Salary
FROM EMPLOYEE
ORDER BY Salary ASC;
```

Retrieve the first name, last name, and birth date of employees, ordered by birth date in descending order (newest first).

```
SQL
SELECT Fname, Lname, Bdate
FROM EMPLOYEE
ORDER BY Bdate DESC;
```

Retrieve the first and last names of employees who work in department number 5 and have a salary greater than 30000.

```
SQL
SELECT Fname, Lname
FROM EMPLOYEE
WHERE Dno = 5 AND Salary > 30000;
```

Retrieve the first and last names of employees who work in department number 4 or have a salary less than 25000.

```
SQL
SELECT Fname, Lname
FROM EMPLOYEE
WHERE Dno = 4 OR Salary < 25000;
```

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Retrieve the first and last names of employees who do not work in department number

```
SQL  
SELECT Fname, Lname  
FROM EMPLOYEE  
WHERE NOT Dno = 1;
```

Retrieve the names of female (Sex = 'F') dependents who are daughters (Relationship = 'Daughter').

```
SQL  
SELECT Dependent_name  
FROM DEPENDENT  
WHERE Sex = 'F' AND Relationship = 'Daughter';
```

Find the department numbers that have projects located in 'Houston'.

```
SQL  
SELECT DISTINCT Dnum  
FROM PROJECT  
WHERE Plocation = 'Houston';
```